Chunder

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CS 401

Introduction to Assembly language

Assembly language is liya perhty h ta ky computer ki working ko greeb sy under stand kr sakain.

For Example:-

C++ apny perhi hoi h.

```
cout << "welcome";
```

is sy kya hota tha Sreen per welcom likha howa a Jata tha

lakin Jo welcom Screen py likh a Jata h. Ya aysy nhi likha

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a Jata.

Computer ky ander mukhtlif kisam ki working hoti processer processing krta h. yani koi working hoti h. asky backend py jo working hoti. wo working jo computer krta h. osko hum assembly language mei understand krain gain.

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Programing languages

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Low level

jis me 0,1 use hota

Machine lang

- esi language hai jo machine ko smj ajati h
- is mai 0,1 use hota h
- very difficult to program

Assembly lang:

esi language h jis mai 0,1 ky sath kuch english ky words b likh sktay h
Yeh machine ko samjh nahi ati - is liy we use translator.

High level

jis me 0,1 use nahi hota.
or jis me english li jati h.

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Assembly language ka
jo Translator hota h. osy
hum assemblor kehty h.
assembler kya krta assembly
language ko Machine
language mai convert krta
h.

High level

C++
C language
Java
C#

ya sahi high level languag
h. in mai codding krna
easy hota h.

computer ko
jo language samj ati wo
Machine language or 0,1
samj ati h. isko matlb high

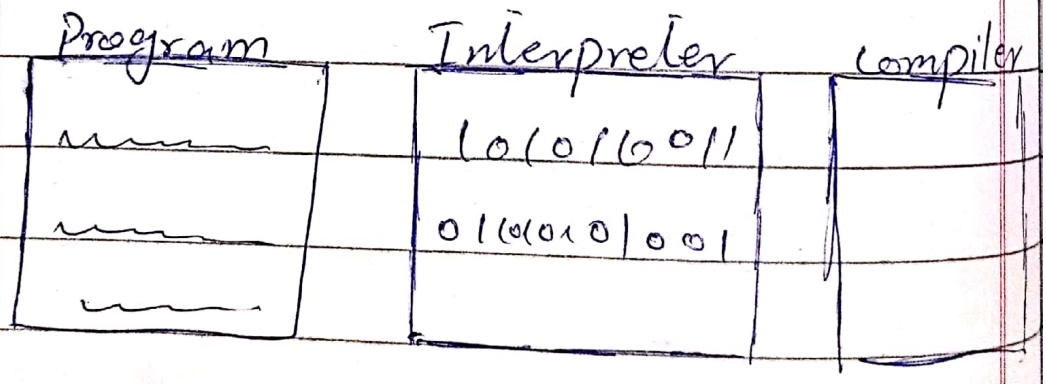
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lvel language computer ko samj nhi aty gi jis mai hum english likhty h. isky 2 Software h. jo high level language ko low level mai convert krty h. yani 0,1 wali language mai

interpreter compiler



Koi program likha h. C++ ka interpreter/compiler kya krta h.

is program ko machin lang mai convert krty h.

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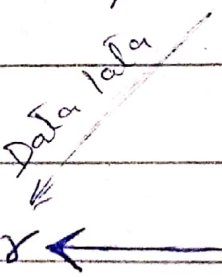
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Interpreter ik ik line leta h. aur isko machine language mai convert krta h.

To code h Translate hokr machine mai ajata h. com jo h osko execute kr dyga.

Compiler kya krta h program ki input ly ga. aur osko akhata generate kr data h

Basic computer Architecture



computer Data lata h aur osko process krta h. aur hamy result dyta h.

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Example:

ik no h oska square hum ny find krna h to number leis mai para hoga RAM ky ander.

user sy data lety h. Tu wo RAM ky ander a Jata h.

RAM ky ander sy data processor mai Jata h. processor os number ka square find kray ga.

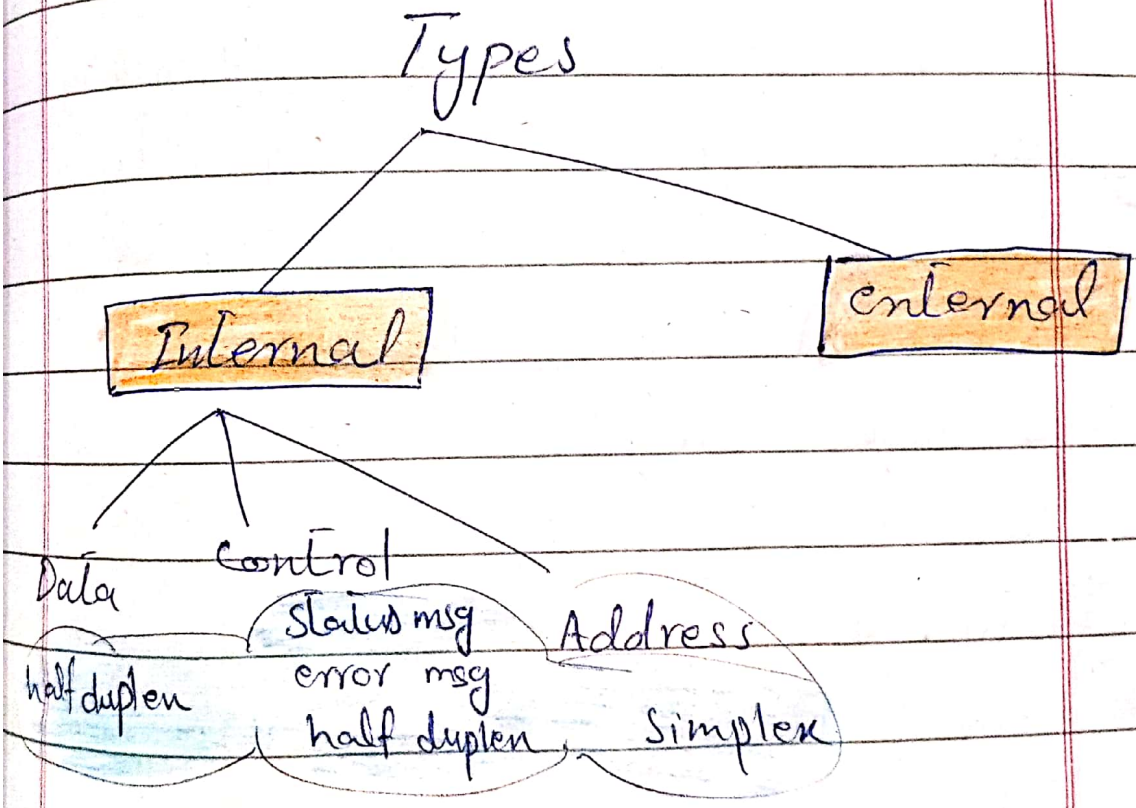
wapis variable ko Ram ky ander slot krwa dyga

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Busses in Assembly Language

Busses are the electrical paths to exchange data from one place to another



2 cheezo ky dermiyan jub communication krnaly h simplex method use krty h is mai kya hota h ky data hmisha 1 direction

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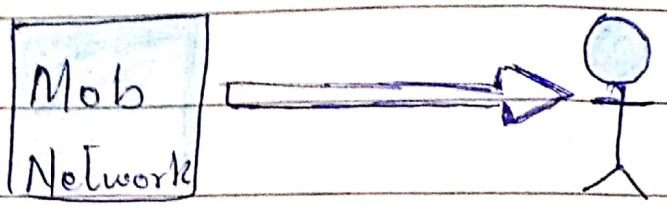
Address bus me hmesha CPU address send krta hai RAM ko... thats x called Simplex.

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mai flow krta h.

Simplex:



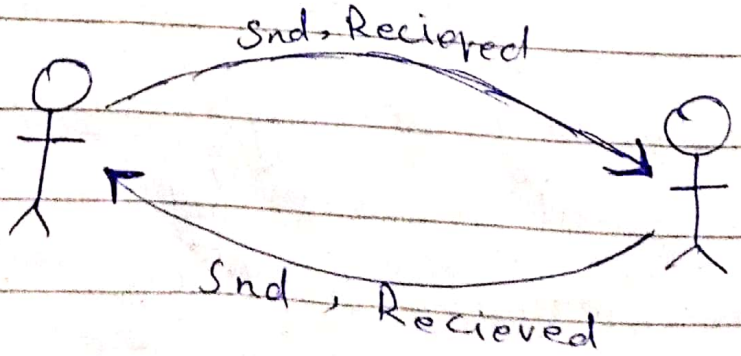
Address bus is one directional.

Sim wally hamy msg krty h. aur hum Recive krty h. is ko hum kehty h.

Simplex Modes.

2nd Method

Half Duplex Mode:.



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is mai ky hota h.
Agr ik bnda data ko snd krta h. Tu dosra snd nhi krsakta.

Agr dosry wala bnda data ko send kr raha h Tu pehly wala data send nhi kr sakta.

Example

Agar kis ny roki Taki ~~tu~~ we kiya ho osmai hota is terha h ky ik bend msg send kr raha hota h Tu dosra oski voice sun sakta h bilkull is terha dosra bend jab button press krta h Tu dosry ko awaz ja rahi hoti h oski.

Jab hum ny processor aur RAM ky dermiyan communication krwani hoti h tu osky

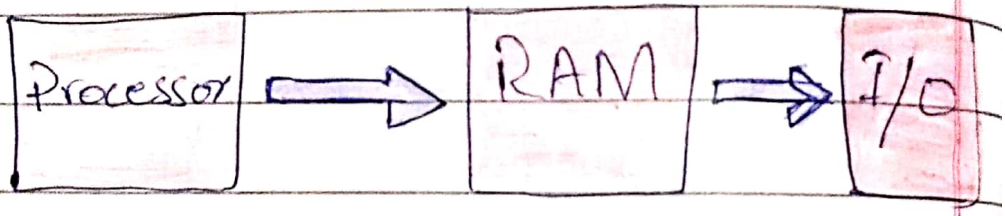
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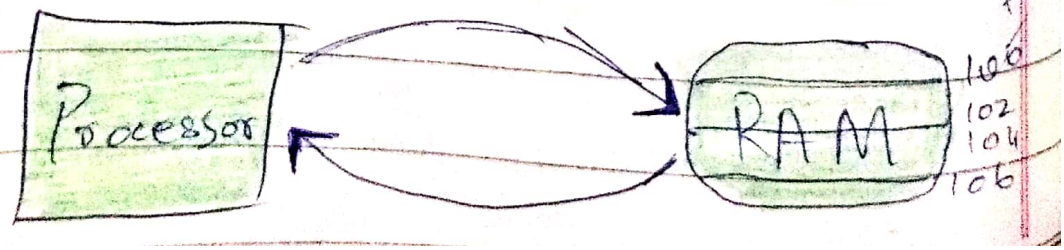
liya internal bussis use hoti h.

aur jub hamy RAM aur I/O devices mai intrection krwani hoti h tu asky liya external bussis use hoti h.



Jo hmara processor hota h wo RAM ky under do operation kr sakta h. RAM ko memory bi kehty h.

RAM data Read krsakta h ya write or stor kr sakta h.



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control bus ko use krty hay Processor RAM ko btata h. ky memry data ko Read krna h. ya phir data ko write krna h.

Agr read ka kehta h tu RAM Phr Tyar ho jati h. ky CPU ny data ko Read krna h.

RAM ky Aander Memory locations bani hoti h. y RAM ky ander memory location bnai hoi h. her memory location ka Adress hota h.

RAM ko ya Tan na hota h ky. konsy Adress ky oper data ko read krna hota h. Tu data ko write krna h.

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Adress is always generated by CPU.

Is osky liy Adress bus use hoti h.

CPU Adress buss ko use krty hoy RAM ko bata h ky humny 100 ko jo memory location ko Read krna hota h.

100 adress py jo data para hota h. Data bus ky zary Data bus ko bejta h.

processor usy processing krta h. Result RAM ky ander oshy stor krwana hota h.

CPU RAM sy data laylta h Seccessfully ik status msg RAM ko genrate krta h osko hum keh sakty h ecknowlege ky data mil gya h.

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as sy RAM ko aram a jata h. Skon a jata h. ☺

kabi kabi processor data access krny ki koshish krta h. RAM ky Ander koi error a jata h. kisi waja sy memory location ko nhi access ker sakta

CPU

is wait ker raha hota h. na ky data ay ga is kam kry ga.

RAM processor ko error msg Bejti h. ya memory location access nhi horahi is mai ya error a raha h.

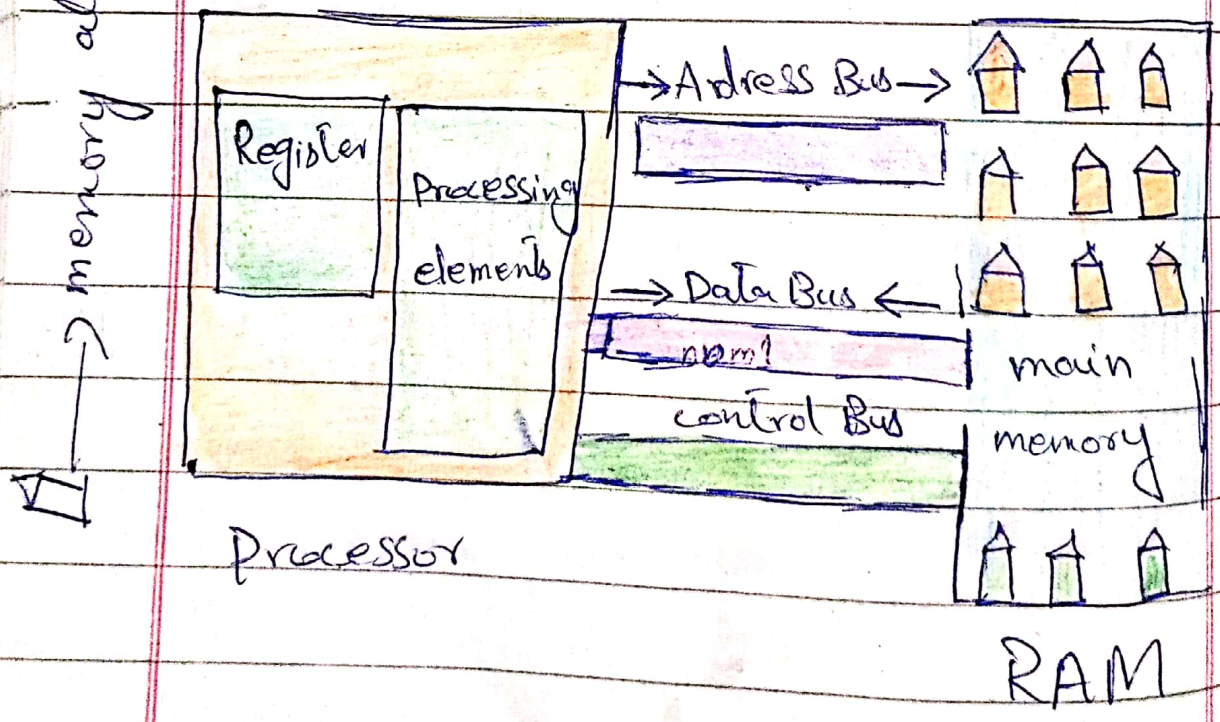
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Registers IN Assembly language

Registers processor ky under holy h:

Register ku zrori h?



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Example: Ya h. do number ka sum calculate krna h sum ky variable mai stor krwana h.

memory allocation locate ho jay gi.

User sy number Inputs lain gain tu memory location mai a jain gain number.

Processor mai 1 number add krny ky bad dosra number add hony hi phla gaib ho jata tu in don numbers ko store krny ky liya

Registers bnay gain. Jab RAM ny phla number diya tha na osy Register ky ander store krwa diya processor ny kaha ky muj

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dosra number do, ku wo ab data bus mai h.

Ab processor key gy ik number Register sy lay gy aur ik data bus sy dono ko add kry gy kisi new memory location mai store kr wa dy gy. Ya Register ka faida thy.

Jis py processor ny processing kirni hoti h.

Registers wo h Jaha hum temporary sur par data stor kr sakty h. ya CPU ky bohot close hoti h. processor ky ander hi hoti h. processing fast hoti h.

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Register ki wja sy processing ku fast hoti h.

Agr processor ny RAM sy data lena h.

control bus sy msg bejna perta h Address generate krna perta h. phir wait ker raha hota h. cpu ky RAM Data send kry gi.

Register sy lany ky liya osy kisi ko cl kry ki zrorat nhi Register osy under hi hota h. waha sy data layta h.

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Types of Register

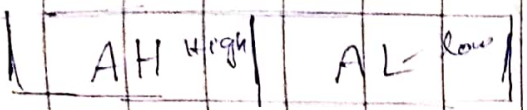
General Purpose

Uses for more then one purpose

1. AX Accumulator
2. BX base
3. CX count
4. DX destination

AX	(16)
EAX	(32)
RAX	(64)

X = Size (16, 32, 64, 80)



Special Purpose

used for only one purpose for which they are made

1. IP (Pc) = Instruction pointer / Program Counter
2. IR = Instruction Register
3. BP = Base Pointer Register
4. SP = Stack pointer register
5. SI = Source index Register
6. DI = Destination Register

C.S = code Segment Register
D.S = Data "
E.S = Extra "
S.S = Stack

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Segment

Code
Data
Extra
Stack

S.S
D.S
E.S
S.S

CPU word Size / CPU width: It means how many no. of bits that can be processed by CPU.

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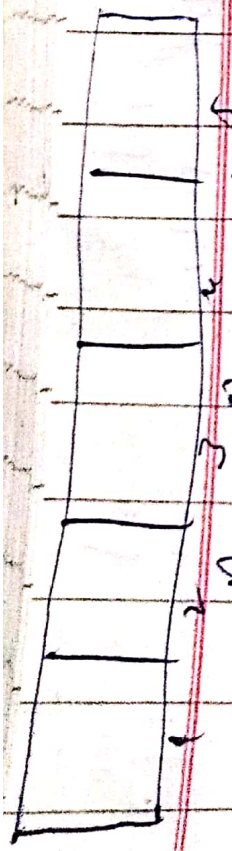
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Jub hum CPU bnany lagty h. Tu os sy phly Accumulator bnaty h.

Jitna Accumulator ka size hota h. utna hi CPU ka size hota h.

Accumulator ka size fine krta h. ky CPU keyword ka size kitna hoga.

Accumulator her operation mai involved hota h.



Base Address =

Base mai array hoti h. Array ka name osky starting point ko show krta h.

Data structure ka starting Address store krwaty h

Base Address = starting Addr

for (int I=0; I<10; i++)
cout << "welcome";

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CX Counter Register

C++ mai aik loop h aur I ik variable h. I zero hoga screen py condition check hogi screen py welcom display hoga.

Pichy sy osy phir I ki value yad nhi rahy gi. Tu osky liya Regis. ik Register hota h. I ki value ko counter kehty h ya ik ik terky count ker raha hota h. Phr Register ky sath match ker ta h.

Jo Increment hota h. wo Register mai 1 add ker data h.

Jo counter ki value hoti h. wo CX ky ander stor hoti h.

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DX :1 Destination:

Ya basically destination key liya use hota h.

final output operation ki store krwa sakty h.

num1 * num2 = 38bits

Accumulator AX 32 bit ka h. ab osky under ya ams save nhi hoga. agye save krny ki koshish krain gain to 32bit ka ho jay ga. baki jo 6 bit h? osky do solutions h.

32 bit store krwaky 6bit zya: kr dain. Data zya tu nhi kr sakty na. osky liya hum kya

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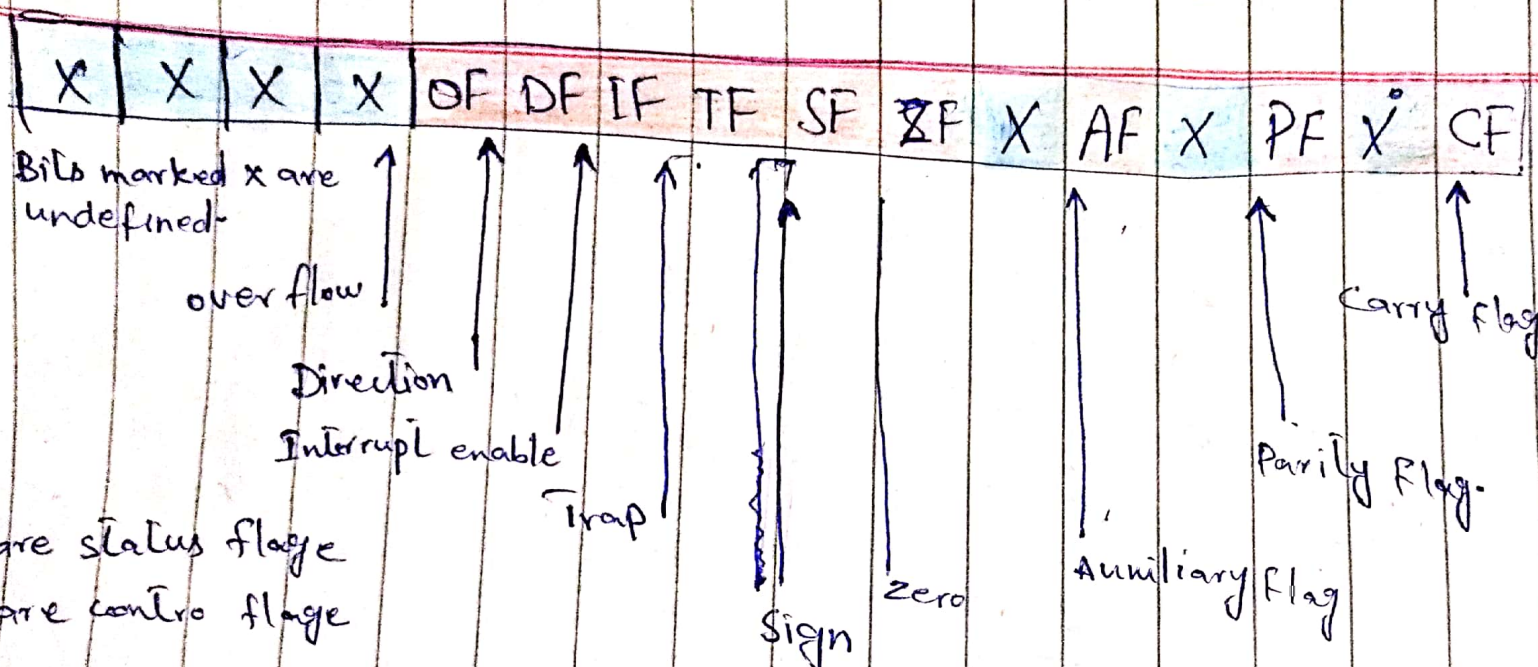
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AX ky ander hum 32 bit store krwa sakty h. aur jo 6 bit bachain gi wo hum DX mai stor krwain gain. in isleia hmara data zya nhi hota.

Ya sari Registers Input leny output leya liya use hoty h.

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8086 Architecture



→ over all microprocessor state represented

→ isley under bits hoti h. her bit ka apna meaning hota h.

→ jiny bi flag h inki do values ho sakti h. Ya one ya 0.

Flag Register
IP Register
Instructions of assembly language -

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Carry flage :-

carry no

① 1100 1000 ax
+ 1101 0010 bx
① 1101 1010

assume 8 bit

Jub hum do number ko add krty h. last py carry ay tu. carry flage ki value 1 ho jati h. ager carry na ay tu carry Flag ki value 0 hoti h.

Carry flag ky do kam hoty h.

Carry flag ki value 1 hoti h. agr ay carry

Agr carry na ay to iski value 0 hogi.

Register ka size 8bit h. Tu 1 ko hum carry flage mei stor krwa dety gain

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data zaya nhi hoga.

Parity flag:

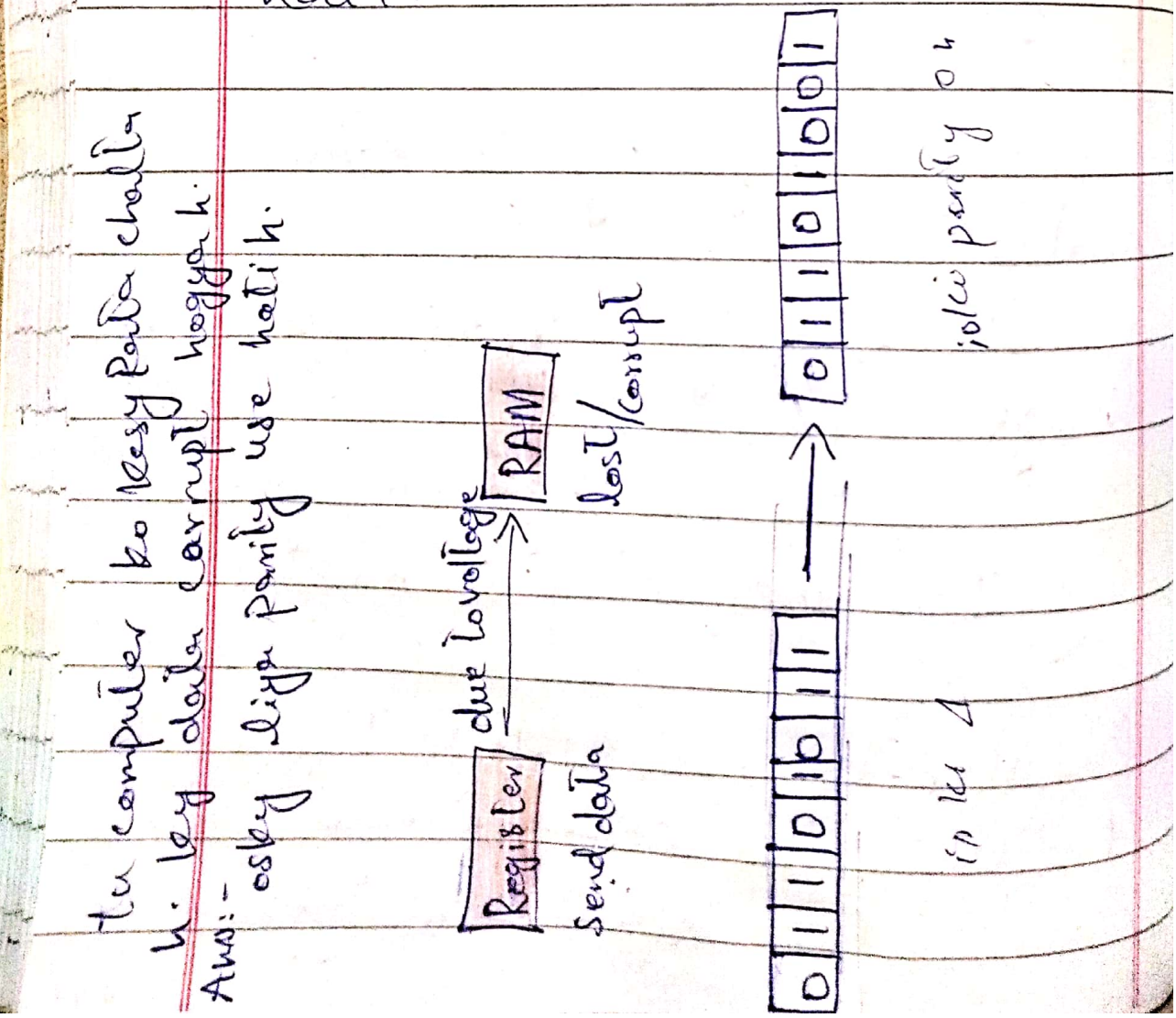
is used for error detection and correction

To ensure that

Jo send kiya tha data wohi receive huwa hai??

tu computer ko kesy pata challa ki kay data corrupt hogya hi.

Ans:- osky liya Parity use hoti hi.



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1k flag Register bnaya.

is mai hum even parity lain gain

even parity :-
if 1's is even = 0
if 1's is odd = 1

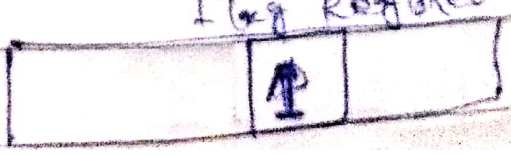
suppose :>

Register sy data send krna h. ku osky liya hum ny even liya h. ku os ky liya number of ones ko count kry ga

jub numbe of odd ku parrity 1 bnti h flage Register ky under parity h wo 1 ho jay gi.

ku kain no of 1 odd ho tu parity 1 hogi

Flag Register



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Auxilary flag:-

Ya bohi kam use hota h

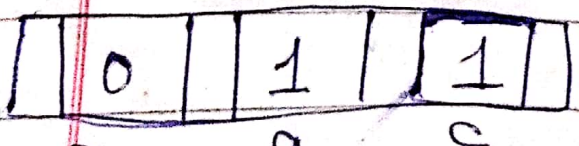
Ya kub 1 bota h?

Jub 3 bit tey bad 1 carry ay tu oski value 1 hojati h.

4bits ka ik group hota h. osy hum nibel kehty h. DRD mai perha hoga.

nibul mai carry ay tu iski vallue 1 hoti h.

last py jub carry ata h tu oski value 1 ho jati h.



① ① ① ①
0100 1100
1100 1110
1 0001 1110

Ku tey Ans nai hoga? Pass oo nhi aya.

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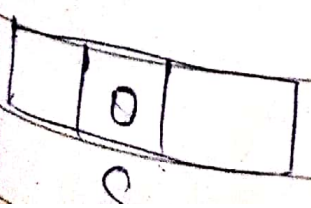
zero flag ki value kab 1 hoti
Agr hum koi operation
krain tu os ki ~~value~~ Ans 0
a jay to tu iski value 1
ho jati h.

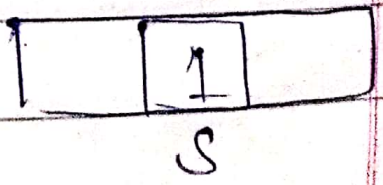
aur answer 0
na aay tu. tu iski value
zero ho jati h.

Sign flag:-

Agr hum
koi bi operation krain
to oski value possitive
ay tu oski value 0 hogi
aur agr Answer negitive
ay tu ans 1 hoga.

$$\begin{array}{r} 2 \\ +\ -6 \\ \hline -4 \end{array} \qquad \begin{array}{r} 6 \\ +\ 2 \\ \hline 8 \end{array}$$





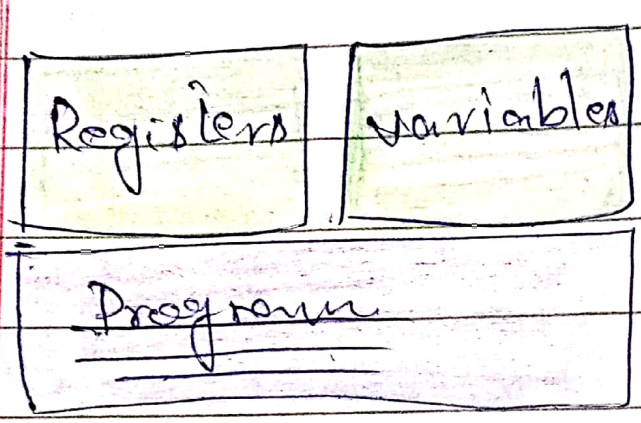
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Trap flag:

basically Debugger ko facilitate krny ky liye use hota h.



Trap flag ki value jab 1 fine hoty h. Single step execution mode mai chala jata h. Yany ik ik line ko execute kry ga. Phr wo bata h. Register mai kya value pary h. variable mai ky value pary h.

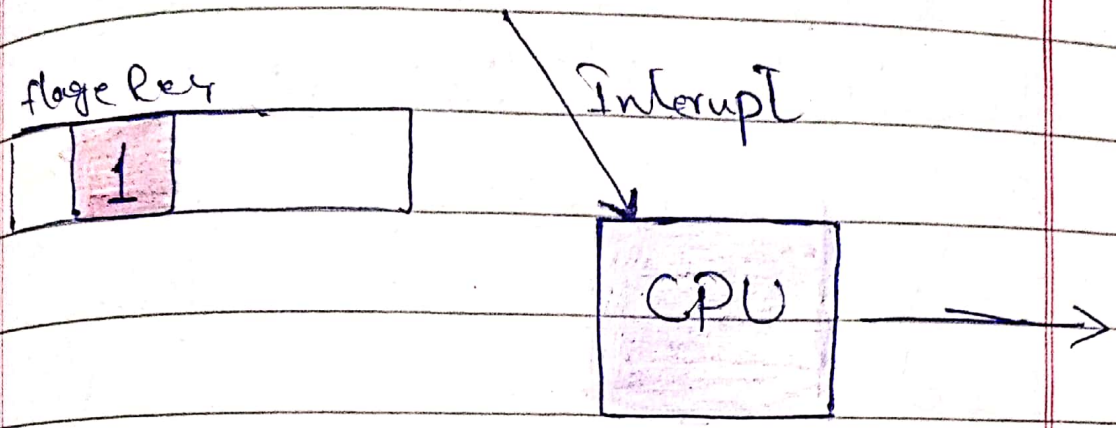
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Interrupt flage :-

a Single generated by h/w or s/w for cpu



CPU data ko processing kr raha tha ilny mai ik Intrupt aya kis software ki taraf sy

iski value agr 1 set h tu apny wala kam chod key Jo Intrupt wala kam aya h iski processing start kr dyay.

Agr value 0 set h.

| 0 |

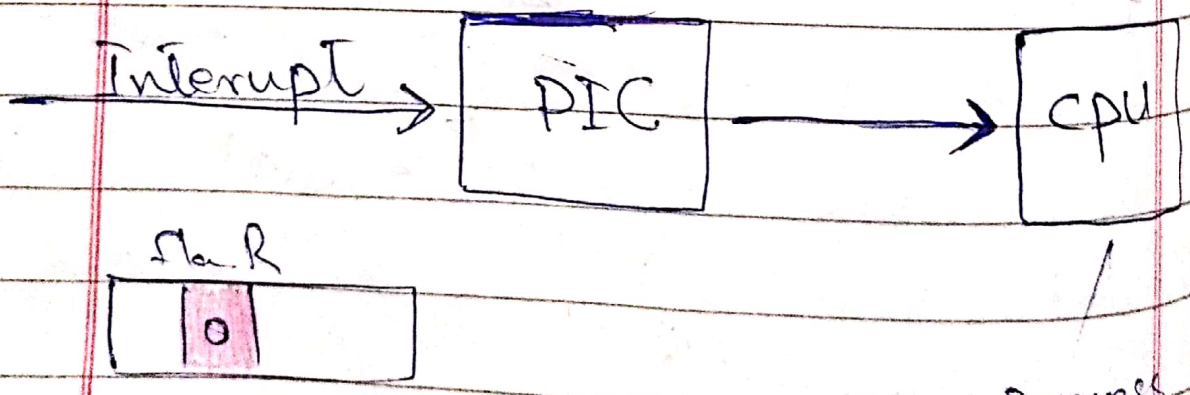
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CPU apni processing kr rahe tha itny mai Interupt aya CPU dakhy ga 0 h. iska matlb h koi bi Interupt, apna kam both Important h. Interupt ko Ignore kr dyga aur CPU apna kam krta rahy ga.

Programmable Interupt Controller



CPU Available nhi h. Process krta h

14 → 13 → 12 → 11 que bna dalyga

Jab cpu free hoga tu yeh ik ik kr key process kry ga.

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Direction flag:-

String ki processing kesy krni h.

Agr Direction flag: ki value zero h tu

"I love Assembly"

aur agar value 1 set krly h.

"ylbmessAevoLi"

Overflow flag:-

n1 * n2 = Answer more then 16 bits

n1 + n2 = 17 bits

Ya hamy bata h ky data zaya ho raha or ak value 1 hai h.

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IP/PC register (Instruction pointer / program counter register)

It holds the Address of the next Instruction to be executed.

IR (Instruction register):-

It holds the Instruction that is in CPU

Instruction Groups:-

Data movment Instruction

destination Register, source.

mov ax, bx mov ax, 9

da 89

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bx mai jo value pari hoi h. osko ax mai move krwana h. 98 ki value ko ax mai dal do.

mov ax, 89

Arithmetic / logic Instructions

2 1 3

Add ax, 33
add bx, [100]
and ax, 23

Sequence learn krna h. Pahli ax mai jo value Pari hoi h. osko add krdo 33 mai aur ans ax ma store krwana h.

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Program control group :-

Cmp ax, 12
jne 100

jne :- compare jump not equal
jne :- zero flag is not equal to 1

Ya wo instrections hoti h. Jo program ki execution ko control krti h.

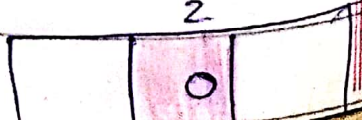
Ya bassically Subtract krti h. Jo flag perhy h. onhy update krti h.

ax mai jo bi value pari h. osko Sub kro 12 ky soth aur Registers ko update krdo

An.

Cmp, 24, 12

Is Answer 0 nhi aya Is zero flage mai value zero a gay gi.



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Cmp 12,12

Answer 0 aya in zero flag me 1

Special Instruction

cli — IntFlag 0 : clear

sti — 1 : set

: interrupt

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Write
assemble
debug
execute

```
[org 0x100]
```

```
mov ax, 0x4c00
Int 0x21
```

Terminate

original

Program

```
[org 0x100]
mov ax, 8
mov bx, 1
add ax, bx
mov bx, 4

mov ax, 0x4c00
int 0x21
```

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dir

Assembler ki 2 types hoti h.

nasm ——— net wise → free hoti h

masm ——— microsoft wala hota h

nasm assembly wali file ko binary code mai convert krta h.

Q

→ Program ko execute kesy krty h.

C:\>nasm first.asm -o fir.com

C:\>fir.com

→ listing file kesy nikalty h

C:\>nasm first.asm -l firstlist.lst

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listing file key ander osny key likha h.

C:\> type firstlist.lst

listing file ko read kesy krty h.

binary

one hexa decimal digit = 4 bits

1 byte = 8 bits

∴ So, 2 hexa digits = 1 byte

hexa

11 = B

14 = E

13 = D

C:\> afd

advance full screen debuger

Sun Mon Tue Wed Thu Fri Sat

back order

Date: 7/20

debuger mai hmein shr .com
ki file open hoti jo bainary
mai assembler ny convert ki
hoti h.

C:\>afd fir.com

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Writ a program in assembly that calculate the sum of Squares of 2 and 3 by adding through registers....

```
[org 0x100]

mov ax, 2
add ax, 2
mov bx, 3
add bx, 3
add bx, 3

mov ax, 0x4c00
Int 0x21
```

mov = 4
add ax, 2
add ax, 2
Square

assignments Question

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Q#1

```
[org 0x100]

mov ax,3
add ax,1
mov ax,2
mov bx,3
sub bx,ax
add ax,bx

mov ax, 0x4c00
int 0x21
```

Question#2

```
[org 0x100]
    mov ax,2
    mov bx,1
    sub ax,bx
    add ax, bx
    mov ax,4
    mov cx,4
```

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```
mov ax, 0x4c00
Int  0x21
```

Question #3

```
[org 0x100]

    mov ax, 2
    mov cx, 4
    mov dx, 1
    add cx, dx
    add ax, dx
    sub cx, dx
    add dx, ax
    mov bx, 8
mov ax, 0x4c00
int 0x21
```

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Memory ko parts mai devide kr ky kesy represent kiya ja sakta h.

Segment Memory Model

Segment = Pieces me divide krna

memory = Jis me data rakhty

Model = Representation

Why ???

- compatability
- Memory Increase

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Compatability

old processors —Program should run on→ new processors

Today's processors

old processors	8088	80X86	80x64
8080, 8bit	16bit	32 bits	64bits

Memory Increase..

48 bit Address

16 bit bus — 16bit address bus — 32 bit Address bus

8 bit data bus — 32 bit data bus

1Kb = 1024 bytes
64Kb = 64*1024 = 65536

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memory mai segmented ram use
hota h.

linear memory Model

8080, 8085 processor thy
inky sath jo Address bus thi
wo 16 bit tei thi.

jitni Address
bus hoti h na process mei
hum 2 ki power lgaty h.

8080, 8085
8bit
16 bit bus

$2^{16} = 65536$
memory location

ik memory location ka jo
Adress hoga wo 16 bit

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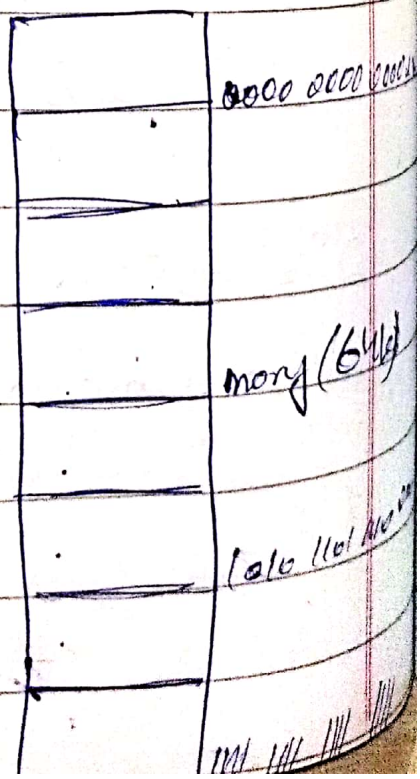
ka hoga. starting h ka phla address h ka sub sy minimum sy start hota h. Minimum bath ka hum 6 bit mai sub ko zero ker dain

every memory location have a address.

last memory ko kesy find kerain.

Sari bits ko hum 1 ker dain jo maximum number ban sakta h wo a jay gy.

8bit ka processor 64k sy zyada memory location cford nhi kerta.



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8080 intel ny launch kiya
Tu oski memory ko Increase
kiya

8085 mai maximum
memory hum liya sakty thy

8080 mai 64kb mai memory
add kei phir 1MB tuk
memory hogi hmari

ab hir memory location ka
jo address hoga 20 bits
ka hoga.

1MB ki memory me hr memory location ka address 20 bit ka hota h.

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8088

16 bit

We devide memory into small pieces of size 64k. These Pieces are called Segments.

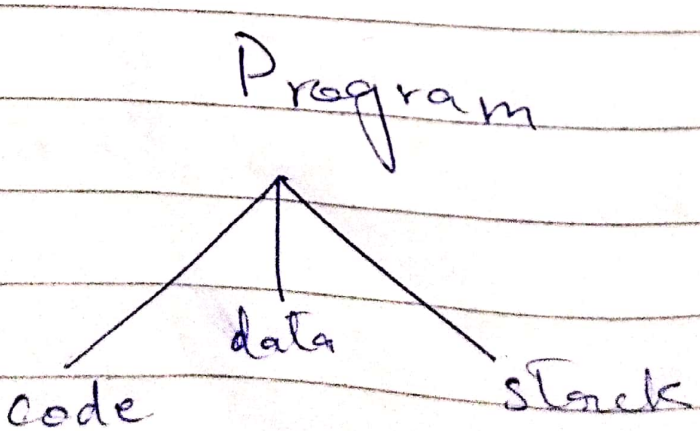
We Increased memory from 64kb to 1MB

every program has a three parts

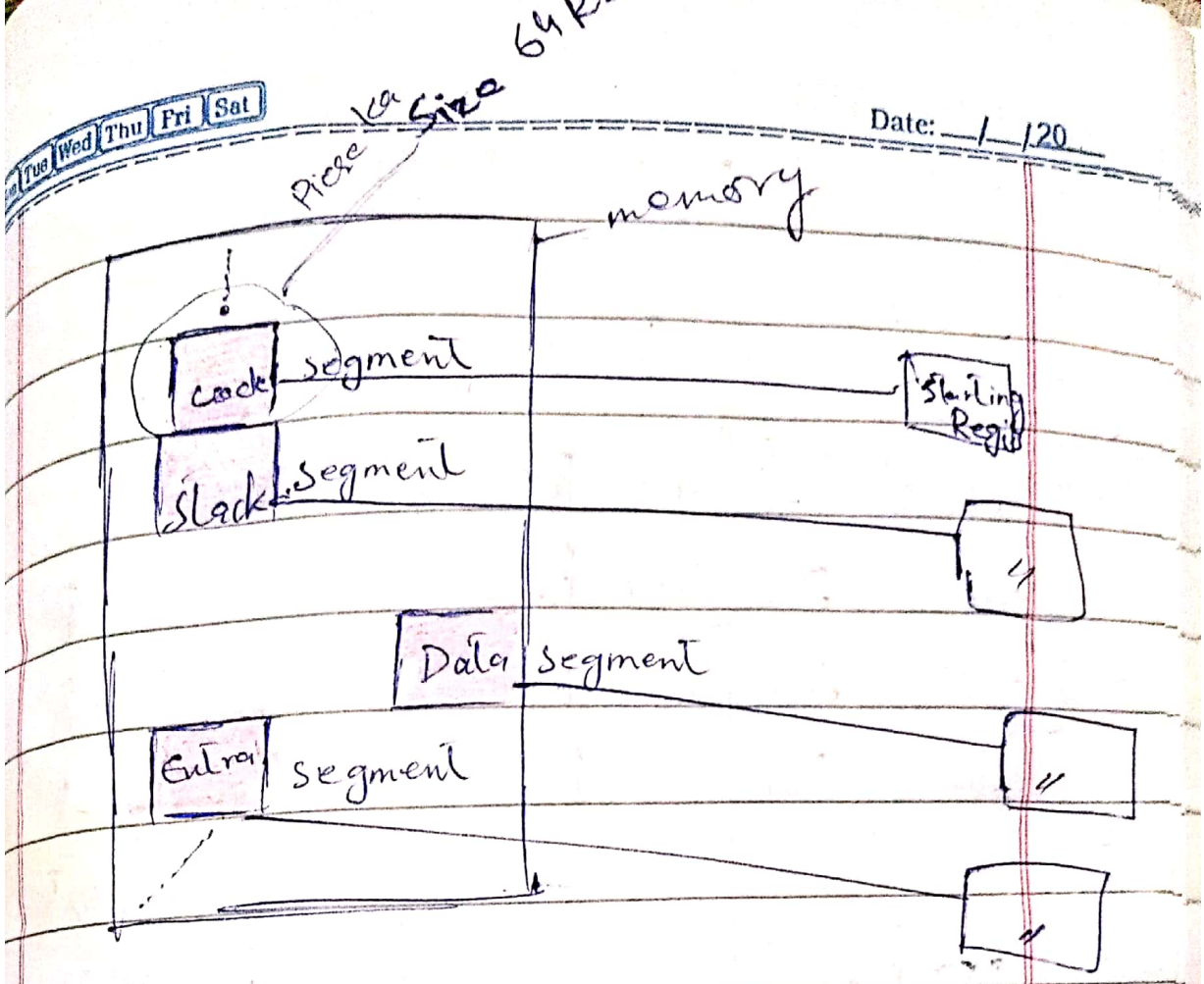
then every memory part devide into pieces.

memory key

her piece ek size 64 ralcha



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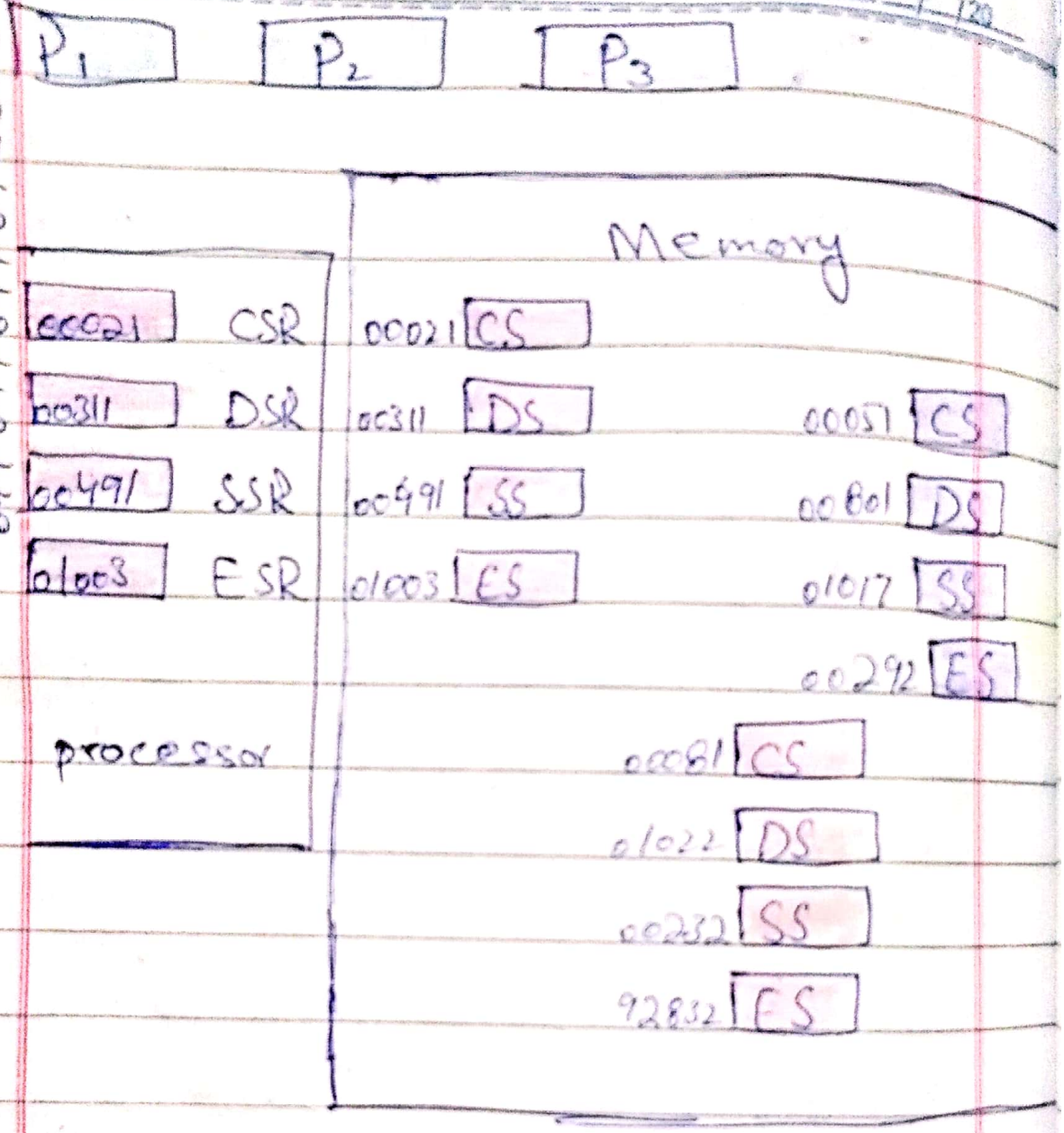


Segment Register:-

CSR:- Stores starting address of code seg.
DSR:- " Data Segment
SSR:- " Stack segment
ESR:- " extra segment.

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4 segments at a time access kr sakty h.
64K + 64 + 64 + 64 = 256K



Program 1 py click krain gai tu wo RAM mai load hogai.

lakin parts mai load ho raha h Segments mai load ho raha h.

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Segment Register mey 00 02 ki values hongi. ky key processor mei phly wala execute ho raha h.

asi lerha p, aur Ps py click kyon.

Segment Registers ki values change hoti jay gi:

Offset Registers

/

Distance

bnay kn gay thy?

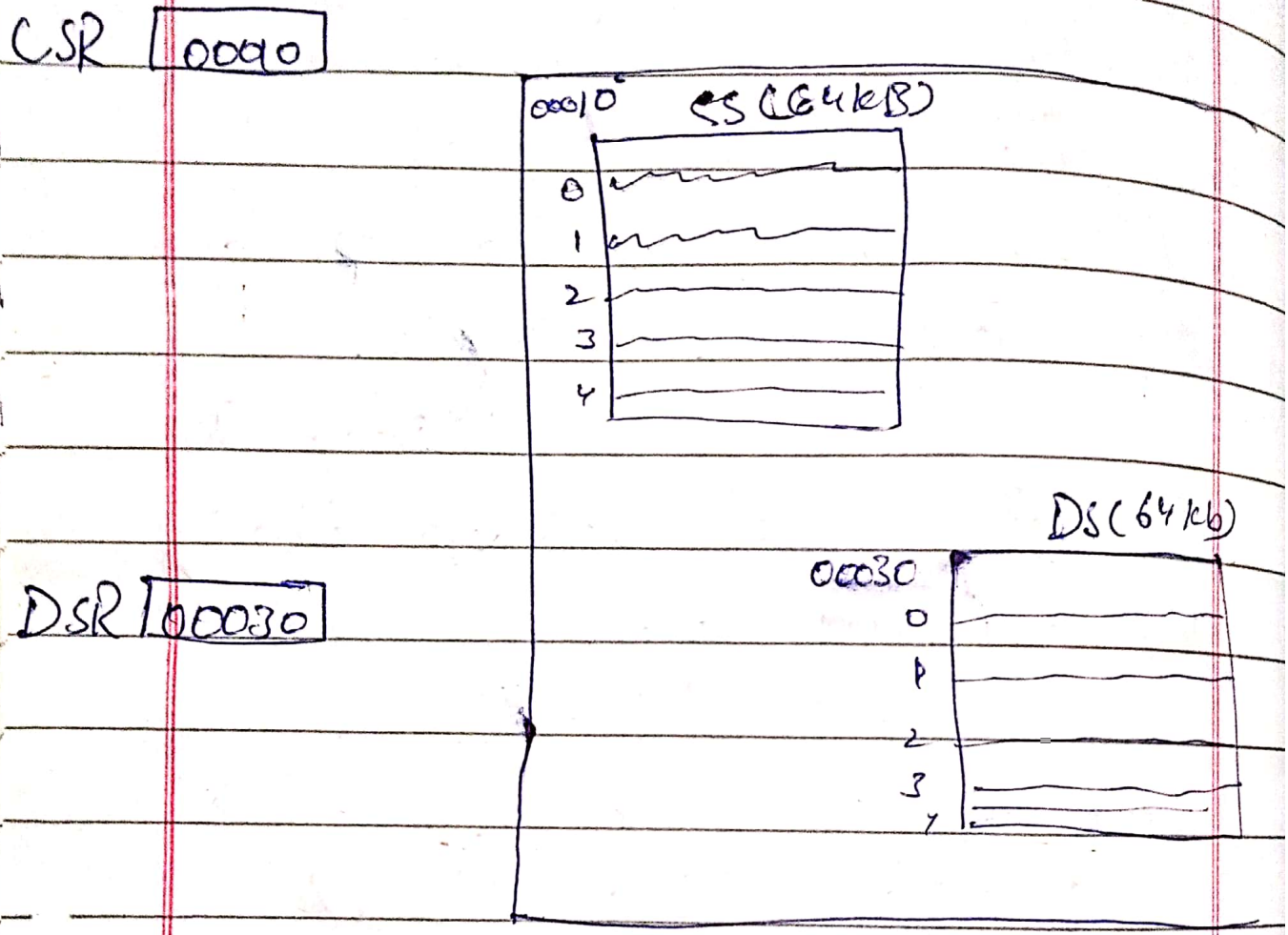
Jis memory location ko access krna h. osky To distance hota h. osko offset kehty h.

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Offset Register



Offset =>

Distance between starting address of segment and desired location.

Code segment ky under memory locations ko numbering dedi Data segment



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